

ξ and Number-Space Dependence in Factorisation

Results documentation of an unresolved working question

Working document. Purpose: preservation of results from an extended investigation without breakthrough. The question was whether the FFGFT parameter $\xi = 4/30000$ shows number-space dependence in factorisation, and whether this follows an FFGFT step hierarchy. All calculations Python-verified. The connection to FFGFT remains a conjecture.

1. Background and Motivation

During work on a T0/FFGFT-based factorisation approach (Python scripts and HTML test programs, 2024–2026), it was observed: the effectiveness of a global ξ -parameter in period-finding (Shor's algorithm) does not appear constant but depends on the number space. Rounding errors and computational precision limits also enter the results indirectly.

Two questions arose: (1) Is there a genuine number-space dependence of ξ in factorisation? (2) Does this follow an FFGFT-type step hierarchy?

Status after extended work without breakthrough: Both questions are neither proved nor disproved. The calculations below provide the established facts.

2. The Algebraically Correct Quantity: $\xi_{\text{num}}(N)$

The HTML file `t0_xi_num_algebraisch.html` correctly defined the key quantity:

$$\xi_{\text{num}}(N) = \lambda(N) / N$$

where $\lambda(N) = \text{lcm}(p-1, q-1)$ is the Carmichael function for $N = p \times q$. This quantity is dimensionless and basis-independent.

Analytical approximation for large primes p, q :

$$\xi_{\text{num}}(N) \approx 1 / (2 \cdot \text{gcd}(p-1, q-1))$$

Key finding (from `t0_primzahl_periodizitaet.html`, Python-confirmed): The period structure of N is determined entirely by $\text{gcd}(p-1, q-1)$. This is a property of the primes themselves — not of a global parameter.

3. Python Verification: ξ_{num} Tables

3.1 Concrete Semiprimes

N	p	q	gcd(p-1,q-1)	$\lambda(N)$	ξ_{num}	Type
15	3	5	2	4	0.2667	Shor classic
35	5	7	2	12	0.3429	
143	11	13	2	60	0.4196	twin primes
65	5	13	4	12	0.1846	gcd=4
301	7	43	6	42	0.1395	gcd=6
77	7	11	2	30	0.3896	gcd=2
4087	61	67	6	660	0.1615	twin ~60
9797	97	101	4	2400	0.2450	smooth p-1
10403	101	103	2	5100	0.4902	twin ~100
291	3	97	2	96	0.3299	very unequal

ξ_{num} range: $[0.14, 0.49] \cdot \xi_{\text{FFGFT}} = 1.333 \times 10^{-4}$ — lies far outside this range.

3.2 Dependence on gcd(p-1, q-1)

gcd	ξ_{num} (analytical)	Factorisability	Example
2	≈ 0.25	hard (RSA class)	twin primes
4	≈ 0.125	moderate	$p \equiv 1, q \equiv 1 \pmod{4}$
6	≈ 0.083	easier	$p \equiv 1, q \equiv 1 \pmod{6}$
12	≈ 0.042	easy	smooth p-1, q-1
30	≈ 0.017	very easy	B-smooth primes
60	≈ 0.008	trivial	highly smooth primes

The number-space dependence is real — but it is classical number theory. Pollard's p-method and Lenstra's ECM exploit precisely this gcd-structure.

4. What the HTML Tests Established

The HTML programs (t0_shor_hypothesis.html, t0_xi_num_algebraisch.html, t0_primzahl_periodizitaet.html, t0_shor_bigint.html) formulated the questions correctly but produced no systematic result tables.

H₀: ξ -optimal bases and random bases produce periods of the same order of magnitude (no advantage from ξ -selection).

H1: ξ -bases produce smaller periods in the median (factor > 1.5 over $N = 50$ trials).

The ξ -resonance function ($\omega=2\pi/r$, $\text{resonance}=\exp(-(\omega-\pi)^2/(4|\xi|))$) is a heuristic without physical derivation. It preferentially selects $r\approx 2$ — the most common gcd-case, not a ξ -effect.

5. Relationship to ξ_{FFGFT}

$$\xi_{\text{FFGFT}} = 1.333 \times 10^{-4} \leq \xi_{\text{num}} \in [0.008, 0.49]$$

The two quantities are not numerically identical and not simply related by a scale factor. A structural connection remains a conjecture without an algebraic bridge proof. What is missing: (1) an argument for why ξ_{FFGFT} (geometric) should appear in ξ_{num} (arithmetic); (2) a scale relation explaining how $\xi_{\text{FFGFT}} \approx 10^{-4}$ maps onto $\xi_{\text{num}} \approx 0.1\text{--}0.5$; (3) a prediction differing from classical number theory.

6. Established Facts — Summary

Statement	Status
$\xi_{\text{num}}(N)$ is number-space dependent	✓ Proved
The dependence is classical number theory	✓ Proved
No global ξ optimises factorisation universally	✓ Proved
ξ -resonance heuristic favours $r\approx 2$	✓ Clear
Rounding errors affect effective ξ	✓ Known
ξ_{num} follows FFGFT step hierarchy	× Not proved
ξ_{FFGFT} and ξ_{num} structurally related	× Not proved
ξ -basis selection improves period-finding	× H_0 not refuted

Conclusion: The number-space dependence is real but classical. The connection to FFGFT geometry is open and remains a conjecture. The breakthrough is missing because the algebraic bridge proof is missing — it would need to show why a geometric invariant of T^4 appears in the Carmichael structure of arithmetic number spaces.